

Conditional Sublogarithmic Approximate-Private Learning from Global Stability

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1 Theoretical Setup

1.1 Problem Setting

Let $\log$ denote the natural logarithm. Let $\log^*$ denote base-two iterated logarithm, with $\log^*(t)$ the least $j \geq 0$ for which the j -fold iterated base-two logarithm of t is at most 1. For a domain X and a nonempty finite binary concept class $C \subseteq \{0, 1\}^X$, set

$$d(C) = \text{VC}(C), \quad \ell(C) = \text{LD}(C), \quad s(C) = 1 + \log^*(1 + \ell(C)), \quad q(C) = d(C) + s(C).$$

Thus $q(C) \geq 1$. Finiteness gives the counting bound

$$\ell(C) \leq \log_2 |C|.$$

Write $Z_X = X \times \{0, 1\}$ and $\mathcal{H}_X = \{0, 1\}^X$. A distribution Q on Z_X is realizable by C if some $c \in C$ satisfies $Q\{(x, y) : y = c(x)\} = 1$. For $h \in \mathcal{H}_X$, define

$$R_Q(h) = \Pr_{(x,y) \sim Q}[h(x) \neq y].$$

All randomized algorithms are kernels on output spaces in which the singleton hypotheses and output events used below are measurable. An equality $G(S) = h$ is equality as functions on X , and its probability includes both the iid sample and all internal randomness.

Two ordered samples of equal size are replacement-adjacent if they differ in at most one record. A kernel $A : Z_X^n \rightsquigarrow \mathcal{H}_X$ is (ε, δ) -differentially private if, for every such S, S' and every measurable output event E ,

$$\Pr[A(S) \in E] \leq e^\varepsilon \Pr[A(S') \in E] + \delta,$$

and the same inequality holds after interchanging S, S' . It is distribution-free realizable (α, β) -PAC for C if, for every Q realizable by C ,

$$\Pr_{S \sim Q^n, h \sim A(S)}[R_Q(h) > \alpha] \leq \beta.$$

There is no properness, output-representation, or computational restriction. Fix

$$\alpha_0 = \frac{1}{8}, \quad \beta_0 = \frac{1}{8},$$

and regard $\varepsilon_0 \in (0, 1)$ as an arbitrary fixed constant.

1.2 Assumption

Assumption 1 (Polynomial global-stability profile). *There is a universal integer $a \geq 1$ such that, for every domain X and every nonempty finite $C \subseteq \{0, 1\}^X$, there are an integer*

$$1 \leq m_C \leq q(C)^a$$

and a single randomized nonprivate producer

$$G_C : Z_X^{m_C} \rightsquigarrow \mathcal{H}_X,$$

chosen independently of the realizable distribution, with the following property. For every distribution Q realizable by C , there is $h_{C,Q} \in \mathcal{H}_X$ such that

$$R_Q(h_{C,Q}) \leq \frac{\alpha_0}{2}, \quad \Pr_{S \sim Q^{m_C}, G_C} [G_C(S) = h_{C,Q}] \geq q(C)^{-a}. \quad (\text{GS})$$

The atom may depend on Q , but G_C, m_C , and a may not. Both the producer and the atom may be improper and computationally unbounded.

This atom condition is the sole nonstandard primitive hypothesis. Private learnability, the stable-selection conversion, its quantitative specialization, and all sequence asymptotics are conclusions to be proved.

1.3 Formalized Goal

Under Assumption 1, fix $\varepsilon_0 \in (0, 1)$. Let $\{X_\kappa\}_{\kappa \in \mathbb{N}}$ be any sequence of domains and, for every κ , let $C_\kappa \subseteq \{0, 1\}^{X_\kappa}$ be a nonempty finite binary concept class. Write $L_\kappa = \log |C_\kappa|$ and use the corresponding subscripts on d, ℓ, s, q . The target is the following conditional statement. Every such domain-and-class sequence satisfying

$$|C_\kappa| \longrightarrow \infty$$

and

$$\forall p \in \mathbb{N} \exists \kappa_0(p) \forall \kappa \geq \kappa_0(p) : \quad L_\kappa > d_\kappa^p \quad (\text{SP})$$

has, at every index, an arbitrary-output, computationally unrestricted, distribution-free realizable (α_0, β_0) -PAC learner that is $(\varepsilon_0, \delta_\kappa)$ -differentially private and uses exactly

$$N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0)$$

records, where the functions N_a and δ_a are defined in Section 2. The required conclusions are

$$N_\kappa = L_\kappa^{o(1)} = o(L_\kappa), \quad \delta_\kappa N_\kappa^\rho \longrightarrow 0 \quad \text{for every fixed } \rho > 0.$$

The same learner must remain eligible whenever an allowed privacy schedule is pointwise no smaller than δ_κ , including the standard allowance $c/[N_\kappa^2 \log(eN_\kappa)]$ eventually for every fixed $c > 0$. Consequently no eventual positive $\Omega(L_\kappa)$ lower bound can hold on this unrestricted learner interface. The conclusion remains conditional on Assumption 1; the assumption itself is not asserted or proved.

2 Preliminaries

For the universal exponent a in Assumption 1, define

$$b = 2a + 2, \quad u(L) = \log \log(e^e + L),$$

$$T_a(q, L) = q^b + u(L), \quad \delta_a(q, L) = e^{-T_a(q, L)}$$

for $q \geq 1$ and $L \geq 0$. These quantities expose separately the polynomial complexity contribution and the slowly growing cardinality correction in the privacy parameter stated below.

For a constant $K_a \geq 2$, to be selected in the proof with dependence only on a, α_0, β_0 and the universal numerical constant in the stable-selection bound, define

$$N_a(q, L, \varepsilon) = \left\lceil K_a \varepsilon^{-2} q^b (1 + T_a(q, L)) \right\rceil.$$

This is the exact fixed input arity used in the main theorem. No class, cardinality, distribution, sequence-index, or ε -dependence is hidden in K_a .

For a class sequence, abbreviate

$$L_\kappa = \log |C_\kappa|, \quad d_\kappa = d(C_\kappa), \quad \ell_\kappa = \ell(C_\kappa), \quad s_\kappa = s(C_\kappa), \quad q_\kappa = q(C_\kappa),$$

$$N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0), \quad \delta_\kappa = \delta_a(q_\kappa, L_\kappa).$$

These abbreviations are used only to state the exact per-class learner parameters and their deterministic sequence limits.

3 Main Theorem

Theorem 3.1 (Conditional sublogarithmic approximate-private learning). *Under Assumption 1, fix $\varepsilon_0 \in (0, 1)$. Let $\{X_\kappa\}_{\kappa \in \mathbb{N}}$ be any sequence of domains and, for every κ , let $C_\kappa \subseteq \{0, 1\}^{X_\kappa}$ be a nonempty finite binary concept class. Suppose that*

$$|C_\kappa| \rightarrow \infty$$

and, for every $p \in \mathbb{N}$, there is $\kappa_0(p)$ for which

$$L_\kappa > d_\kappa^p \quad (\kappa \geq \kappa_0(p)).$$

There is a constant $K_a \geq 2$, depending only on a, α_0, β_0 and the universal numerical constant in the stable-selection bound, and independent of $\kappa, X_\kappa, C_\kappa, |C_\kappa|$, every realizable distribution, and ε_0 , such that the following holds. For every κ , there is an arbitrary-output, computationally unrestricted kernel

$$M_\kappa : Z_{X_\kappa}^{N_\kappa} \rightsquigarrow \mathcal{H}_{X_\kappa}$$

that is distribution-free realizable (α_0, β_0) -PAC in population zero-one risk and $(\varepsilon_0, \delta_\kappa)$ -differentially private under symmetric ordered replacement adjacency, where

$$\delta_\kappa = \exp[-q_\kappa^{2a+2} - \log \log(e^e + L_\kappa)],$$

$$\log \frac{1}{\delta_\kappa} = q_\kappa^{2a+2} + \log \log(e^e + L_\kappa),$$

and

$$N_\kappa = \left\lceil K_a \varepsilon_0^{-2} q_\kappa^{2a+2} (1 + q_\kappa^{2a+2} + \log \log(e^e + L_\kappa)) \right\rceil$$

$$\leq 1 + K_a \varepsilon_0^{-2} q_\kappa^{2a+2} (1 + q_\kappa^{2a+2} + \log \log(e^e + L_\kappa)).$$

The probability in the PAC statement is joint over the iid sample and all learner randomness; privacy is pointwise for every adjacent pair and measurable output event. The input arity is exactly N_κ at each class, while the sequence-asymptotic conclusions are

$$N_\kappa = L_\kappa^{o(1)} = o(L_\kappa)$$

and, for every fixed real $\rho > 0$,

$$\delta_\kappa N_\kappa^\rho \longrightarrow 0.$$

If an allowed schedule satisfies $\bar{\delta}(N_\kappa) \geq \delta_\kappa$, then the same learner is $(\varepsilon_0, \bar{\delta}(N_\kappa))$ -differentially private. In particular, for every fixed $c > 0$, eventually

$$\delta_\kappa \leq \frac{c}{N_\kappa^2 \log(eN_\kappa)}.$$

At the exact schedule or any eventually larger allowed schedule, there are no $c_* > 0$ and κ_* such that every eligible arbitrary-output, computationally unrestricted, distribution-free realizable (α_0, β_0) -PAC learner for C_κ uses at least $c_* L_\kappa$ records for every $\kappa \geq \kappa_*$.

All exposed variables are displayed above. The only hidden numerical dependence is the stated dependence of K_a ; a , $\alpha_0, \beta_0, \varepsilon_0$, the sequence, the zero-one risk, and the adjacency convention are fixed while $\kappa \rightarrow \infty$. This is a conditional result: it assumes, and does not prove, Assumption 1.

4 Proof Sketch

The proof first conjugates binary labels and hypotheses by the pointwise bijection $y \mapsto 2y - 1$. This preserves realizability, zero-one loss, iid laws, exact atom probabilities, measurable output events, and replacement adjacency. The stable-selection theorem of [Bun et al. \(2021\)](#) can therefore be applied to the producer in Assumption 1. Its construction depends on the fixed producer and public parameters, not on the realizable distribution or its frequent atom, which yields one distribution-free arbitrary-output learner.

The resulting two-term sample bound is then specialized without discarding any logarithmic or confidence term. With $b = 2a + 2$, every term is dominated explicitly by $q^b(1 + T_a(q, L))$, and a constant K_a with only the declared dependence suffices. Prefix projection pads the learner to exactly $N_a(q, L, \varepsilon_0)$ inputs while preserving both utility and privacy.

Next, the privacy parameter is evaluated exactly:

$$\delta_a(q, L) = \frac{e^{-q^b}}{\log(e^e + L)}, \quad \log(1/\delta_a(q, L)) = q^b + \log \log(e^e + L).$$

The cardinality factor remains active when q is bounded. For a class sequence, the full quantifier order in the superpolynomial condition and $\ell_\kappa \leq L_\kappa / \log 2$ imply $q_\kappa = L_\kappa^{o(1)}$. Direct exponent allocation through T_a , the product defining N_a , and its ceiling then proves $N_\kappa = L_\kappa^{o(1)} = o(L_\kappa)$.

Finally, writing $x_\kappa = q_\kappa^b$ and $u_\kappa = u(L_\kappa)$, the exact privacy product factors into an x_κ -term that is uniformly bounded on $[1, \infty)$ and an u_κ -term that tends to zero:

$$\delta_\kappa N_\kappa^\rho \leq C_{\rho, \varepsilon_0, K_a} [e^{-x_\kappa} x_\kappa^\rho (1 + x_\kappa)^\rho] [e^{-u_\kappa} (1 + u_\kappa)^\rho].$$

This proves privacy negligibility for every fixed $\rho > 0$, including bounded q_κ . Privacy monotonicity is used only toward a larger additive allowance, and the exact learners with $N_\kappa / L_\kappa \rightarrow 0$ contradict every eventual positive linear lower bound. The complete derivations and the final assembly appear in Section A.

References

BUN, M., LIVNI, R. and MORAN, S. (2021). An equivalence between private classification and online prediction. Version 3, revised 22 June 2021. [4](#), [7](#)

A Proof Details

A.1 Stable Selection Under Binary Recoding

Lemma A.1 (Exact measurable binary recoding). *Let X be a domain, let $r \geq 1$, let $C \subseteq \mathcal{H}_X$, let Q be a distribution on $Z_X = X \times \{0, 1\}$, let $h \in \mathcal{H}_X$, and let $G : Z_X^r \rightsquigarrow \mathcal{H}_X$ be a randomized kernel. Then the pointwise bijection between the labels $\{0, 1\}$ and $\{-1, +1\}$, equipped with the transported output sigma-algebra, preserves realizability, population zero-one risk, iid product laws, exact output-atom probabilities, and ordered replacement adjacency in both directions.*

Proof. Define

$$\lambda(0) = -1, \quad \lambda(1) = +1, \quad \lambda^{-1}(t) = \frac{t+1}{2},$$

and

$$\Phi_Z(x, y) = (x, \lambda(y)), \quad \Phi_H(h)(x) = \lambda(h(x)).$$

Both maps are pointwise bijections. If Σ_X is the output sigma-algebra on $\mathcal{H}_X$, equip $\mathcal{H}_X^\pm = \{-1, +1\}^X$ with

$$\Sigma_X^\pm = \{\Phi_H(E) : E \in \Sigma_X\}.$$

Because Φ_H is bijective, $\Sigma_X^\pm$ is a sigma-algebra and both Φ_H and Φ_H^{-1} are measurable. In particular, measurable singleton hypotheses remain measurable. On labeled records we use the finite discrete label sigma-algebras and the corresponding recordwise transport. The standing measurability convention therefore covers the equality, histogram, singleton, and output events used below.

Set

$$Q^\pm = (\Phi_Z)_\# Q, \quad G^\pm(S^\pm) = \Phi_H(G((\Phi_Z^{-1})^{\otimes r}(S^\pm))). \quad (\text{A.1})$$

The transported structures make $G^\pm$ a kernel. If Q is realizable by $c \in C$, then for $Q^\pm$ -almost every (x, t) ,

$$t = \lambda(c(x)) = \Phi_H(c)(x),$$

so $Q^\pm$ is realizable by $\Phi_H(c) \in C^\pm := \Phi_H(C)$.

For every $h \in \mathcal{H}_X$, pointwise bijectivity gives

$$\begin{aligned} \text{loss}_{Q^\pm}(\Phi_H(h)) &= \Pr_{(x,t) \sim Q^\pm} [\Phi_H(h)(x) \neq t] \\ &= \Pr_{(x,y) \sim Q} [\lambda(h(x)) \neq \lambda(y)] \\ &= \Pr_{(x,y) \sim Q} [h(x) \neq y] = R_Q(h). \end{aligned} \quad (\text{A.2})$$

The recordwise map gives the exact product-law identity

$$(Q^\pm)^r = (\Phi_Z^{\otimes r})_\# Q^r. \quad (\text{A.3})$$

Combining (A.1), (A.3), and bijectivity of Φ_H , including all internal randomness of G , yields

$$\Pr_{S^\pm \sim (Q^\pm)^r, G^\pm} [G^\pm(S^\pm) = \Phi_H(h)] = \Pr_{S \sim Q^r, G} [G(S) = h]. \quad (\text{A.4})$$

Finally, Φ_Z is a bijection on individual records. Hence for ordered samples S, S' of the same length,

$$\#\{i : S_i \neq S'_i\} = \#\{i : \Phi_Z(S_i) \neq \Phi_Z(S'_i)\}. \quad (\text{A.5})$$

Thus replacement of an instance, a label, or both components of one record is carried to exactly one source-record replacement, and conversely. Equations (A.2)–(A.5) establish all asserted zero-residual correspondences. $\square$

Theorem A.1 (Stable selection from a frequent accurate hypothesis). *Let X be a domain, let $\mathcal{F} \subseteq \{-1, +1\}^X$, let D be realizable by $\mathcal{F}$, let $m \geq 1$, let $\eta \in (0, 1]$, and let*

$$G : (X \times \{-1, +1\})^m \rightsquigarrow \{-1, +1\}^X$$

be randomized. Suppose that some $h \in \{-1, +1\}^X$ satisfies

$$\Pr_{S \sim D^m, G}[G(S) = h] \geq \eta, \quad \text{loss}_D(h) \leq \frac{\alpha_0}{2},$$

where $\alpha_0 = \beta_0 = 1/8$. For every $\varepsilon_0 \in (0, 1)$ and $\delta \in (0, e^{-2}]$, the Bun-Livni-Moran construction gives an integer $n \geq 1$ and an arbitrary-output (ε_0, δ) -differentially private kernel $M : (X \times \{-1, +1\})^n \rightsquigarrow \{-1, +1\}^X$ such that

$$\Pr_{S \sim D^n, M}[\text{loss}_D(M(S)) > \alpha_0] \leq \beta_0$$

and

$$n \leq C_{\text{BLM}} \left[\frac{m}{\eta \varepsilon_0} \log \left(\frac{1}{\eta \beta_0 \delta} \right) + \frac{1}{\alpha_0 \varepsilon_0} \log \left(\frac{1}{\eta \beta_0} \right) \right] \quad (\text{A.6})$$

for a universal numerical constant $C_{\text{BLM}} > 0$.

Source justification. This is Theorem 17 in Section 5.2 of Bun et al. (2021). Its stable-histogram and private selection construction partitions the input into disjoint m -record batches, applies the fixed G to each batch, runs a stable histogram on the resulting hypotheses, and then runs a generic private learner on a fresh labeled subsample using the resulting finite list. The construction depends on $G, m, \eta, \alpha_0, \beta_0, \varepsilon_0, \delta$, but not on D or the witness h ; the latter appears only in the utility analysis. Its codomain is the full $\{-1, +1\}^X$, and its privacy convention is symmetric fixed-length one-record replacement. The source writes (A.6) with absolute $O(\cdot)$ constants in its general parameter statement. In the fixed regime above, $\beta_0 = 1/8$ makes both displayed logarithms at least $\log 8$, and $\delta \leq e^{-2}$ only increases the first one. A single universal C_{BLM} therefore absorbs the source's absolute implementation constants and additive integer choices in precisely this consumed regime, without asserting a pure constant inequality on the unrestricted parameter cube. $\square$

Lemma A.2 (Admissibility of stable selection). *Under Assumption 1, fix a domain X and a nonempty finite $C \subseteq \{0, 1\}^X$, put $q = q(C)$ and $L = \log |C|$, and fix $\varepsilon_0 \in (0, 1)$. Let*

$$\eta_C = q^{-a}, \quad \delta_C = \delta_a(q, L) = e^{-T_a(q, L)}.$$

For every distribution Q realizable by C , the recoded class $C^\pm$, the fixed producer $G_C^\pm$, and the recoded witness $h_{C, Q}^\pm$ satisfy the hypotheses of Theorem A.1 at

$$(m, \eta, \varepsilon_0, \delta) = (m_C, \eta_C, \varepsilon_0, \delta_C).$$

If $q = 1$, then $m_C = 1$ and $\eta_C = 1$, and the same application remains admissible.

Proof. Assumption 1 supplies one integer m_C and one randomized kernel G_C , both fixed before Q , such that

$$1 \leq m_C \leq q^a. \quad (\text{A.7})$$

For every realizable Q , it supplies an arbitrary $h_{C, Q} \in \mathcal{H}_X$ satisfying

$$R_Q(h_{C, Q}) \leq \frac{\alpha_0}{2}, \quad \Pr_{S \sim Q^{m_C}, G_C}[G_C(S) = h_{C, Q}] \geq q^{-a} = \eta_C. \quad (\text{A.8})$$

Apply Lemma A.1 with $r = m_C$. Equations (A.2) and (A.4) turn (A.8) into

$$\text{loss}_{Q^\pm}(h_{C,Q}^\pm) \leq \frac{\alpha_0}{2}, \quad \Pr[G_C^\pm((Q^\pm)^{m_C}) = h_{C,Q}^\pm] \geq \eta_C. \quad (\text{A.9})$$

The same lemma proves that $Q^\pm$ is realizable by $C^\pm$. Neither the witness nor the producer output is required to lie in $C^\pm$.

Since $q \geq 1$ and $a \geq 1$, $0 < \eta_C \leq 1$. The values $\alpha_0, \beta_0, \varepsilon_0$ lie in $(0, 1)$. Moreover, $q^b \geq 1$ and $u(L) \geq 1$, so

$$T_a(q, L) \geq 2, \quad 0 < \delta_C = e^{-T_a(q, L)} \leq e^{-2} < 1. \quad (\text{A.10})$$

At $q = 1$, (A.7) forces $m_C = 1$, while $\eta_C = 1^{-a} = 1$. The logarithms and denominators in (A.6) remain legal because $\beta_0, \delta_C \in (0, 1)$. Thus no $q > 1$ restriction has been introduced. $\square$

Proposition A.1 (A distribution-free source-label learner). *Under Assumption 1, Lemma A.1, Lemma A.2, and Theorem A.1, fix a nonempty finite C , where $C \subseteq \{0, 1\}^X$ for a domain X , put $q = q(C)$, $L = \log |C|$, $\delta_C = \delta_a(q, L)$, and fix $\varepsilon_0 \in (0, 1)$. Then there are one integer n_C and one kernel*

$$A_C^\pm : (X \times \{-1, +1\})^{n_C} \rightsquigarrow \{-1, +1\}^X,$$

both independent of the realizable distribution, such that for every realizable Q ,

$$\Pr_{S^\pm \sim (Q^\pm)^{n_C}, A_C^\pm} [\text{loss}_{Q^\pm}(A_C^\pm(S^\pm)) > \alpha_0] \leq \beta_0, \quad (\text{A.11})$$

$A_C^\pm$ is $(\varepsilon_0, \delta_C)$ -differentially private, and

$$n_C \leq C_{\text{BLM}} \left[\frac{m_C}{q^{-a}\varepsilon_0} \log \left(\frac{1}{q^{-a}\beta_0\delta_C} \right) + \frac{1}{\alpha_0\varepsilon_0} \log \left(\frac{1}{q^{-a}\beta_0} \right) \right]. \quad (\text{A.12})$$

The output range is all of $\{-1, +1\}^X$.

Proof. Fix $C, G_C^\pm, m_C$, and the public parameter tuple from Lemma A.2. Fix also the stable-histogram and private-selection implementations from Theorem A.1. Their construction invokes only the fixed $G_C^\pm$ and the public tuple

$$(m_C, q^{-a}, \alpha_0, \beta_0, \varepsilon_0, \delta_C).$$

Neither $Q^\pm$ nor $h_{C,Q}^\pm$ is an algorithm input or implementation choice. Thus the prescription defines one kernel $A_C^\pm$ and one integer n_C , independent of Q . For each realizable Q , Lemma A.2 supplies a witness for that same fixed kernel. Applying its utility conclusion separately to each Q gives (A.11), with the quantifier order

$$\exists A_C^\pm \quad \forall Q \text{ realizable by } C.$$

Its privacy proof is distribution-independent and supplies both ordered replacement inequalities for every adjacent input pair. Instantiating (A.6) gives (A.12), with the same universal C_{BLM} for all X, C, Q, m_C, q, L and all admissible ε_0 . Both terms remain intact: this argument has not replaced m_C by q^a , expanded the logarithms, dominated the α_0^{-1} term, selected K_a , or invoked a sequence limit. $\square$

Proposition A.2 (Decoded arbitrary-output private learner). *Under Assumption 1, Lemma A.1, and Proposition A.1, fix a domain X , a nonempty finite $C \subseteq \{0, 1\}^X$, and $\varepsilon_0 \in (0, 1)$. Put $q = q(C)$, $L = \log |C|$, and $\delta_C = \delta_a(q, L)$, and let $n_C, A_C^\pm$ be furnished by Proposition A.1. Define, for $S \in Z_X^{n_C}$,*

$$A_C(S) = \Phi_H^{-1}(A_C^\pm(\Phi_Z^{\otimes n_C}(S))). \quad (\text{A.13})$$

Then $A_C : Z_X^{n_C} \rightsquigarrow \mathcal{H}_X$ is one arbitrary-output, distribution-free realizable (α_0, β_0) -PAC learner for C , is $(\varepsilon_0, \delta_C)$ -differentially private under ordered replacement adjacency, and satisfies the sample bound (A.12).

Proof. The maps in (A.13) are bimeasurable by Lemma A.1, so A_C is a kernel fixed independently of Q . If Q is realizable by C and $S \sim Q^{n_C}$, then $\Phi_Z^{\otimes n_C}(S) \sim (Q^\pm)^{n_C}$. For every source output $g^\pm$, (A.2) gives

$$R_Q(\Phi_H^{-1}(g^\pm)) = \text{loss}_{Q^\pm}(g^\pm).$$

Consequently (A.11) implies

$$\Pr_{S \sim Q^{n_C}, A_C} [R_Q(A_C(S)) > \alpha_0] \leq \beta_0. \quad (\text{A.14})$$

The same A_C satisfies (A.14) for every realizable Q , and the probability includes both sample and learner randomness.

For privacy, let $S, S' \in Z_X^{n_C}$ be replacement-adjacent and let $E \subseteq \mathcal{H}_X$ be measurable. By Lemma A.1, $\Phi_Z^{\otimes n_C}(S)$ and $\Phi_Z^{\otimes n_C}(S')$ are source-adjacent and $E^\pm := \Phi_H(E)$ is measurable. Hence

$$\begin{aligned} \Pr[A_C(S) \in E] &= \Pr[A_C^\pm(\Phi_Z^{\otimes n_C}(S)) \in E^\pm] \\ &\leq e^{\varepsilon_0} \Pr[A_C^\pm(\Phi_Z^{\otimes n_C}(S')) \in E^\pm] + \delta_C \\ &= e^{\varepsilon_0} \Pr[A_C(S') \in E] + \delta_C. \end{aligned} \quad (\text{A.15})$$

Interchanging S, S' gives the reverse inequality. Identical samples are immediate, while (A.5) covers replacement of the instance, label, or both components of one record.

The inverse output map is deterministic postprocessing whose image is all of $\mathcal{H}_X$. Thus arbitrary improper outputs and the lack of representation or computational restrictions are preserved, and the sample bound (A.12) is unchanged. Together, Lemma A.1, Lemma A.2, Proposition A.1, and the present argument derive the complete stable-selection conclusion without invoking any later asymptotic specialization. $\square$

A.2 Explicit Sample Domination and Exact Padding

Lemma A.3 (Explicit stable-selection specialization). *Under Assumption 1 and Proposition A.2, fix a domain X , a nonempty finite $C \subseteq \{0, 1\}^X$, put $q = q(C)$, $L = \log |C|$, and fix $\varepsilon_0 \in (0, 1)$. Define*

$$B_0 = \log \frac{1}{\beta_0} = \log 8, \quad D_a = 1 + a + B_0 + \alpha_0^{-1}(a + B_0) = 1 + 9a + 9 \log 8, \quad (\text{A.16})$$

and choose

$$K_a = \max\{2, C_{\text{BLM}} D_a\}. \quad (\text{A.17})$$

If n_C is the sample size in Proposition A.2, then

$$\begin{aligned} n_C &\leq \frac{C_{\text{BLM}}}{\varepsilon_0} \left[q^{2a} \left(a \log q + \log \frac{1}{\beta_0} + T_a(q, L) \right) + \frac{1}{\alpha_0} \left(a \log q + \log \frac{1}{\beta_0} \right) \right] \\ &\leq K_a \varepsilon_0^{-2} q^b (1 + T_a(q, L)) \leq N_a(q, L, \varepsilon_0). \end{aligned} \quad (\text{A.18})$$

The constant K_a depends only on $a, \alpha_0, \beta_0, C_{\text{BLM}}$, and the inequalities remain valid at $q = 1$ and uniformly over $0 < \varepsilon_0 < 1$.

Proof. Write $T = T_a(q, L)$ and $\delta_C = \delta_a(q, L) = e^{-T}$. From Assumption 1,

$$\frac{m_C}{q^{-a}} = m_C q^a \leq q^{2a}. \quad (\text{A.19})$$

The two logarithms in (A.12) expand exactly as

$$\log\left(\frac{1}{q^{-a}\beta_0\delta_C}\right) = \log\left(\frac{q^a e^T}{\beta_0}\right) = a \log q + B_0 + T, \quad (\text{A.20})$$

$$\log\left(\frac{1}{q^{-a}\beta_0}\right) = \log\left(\frac{q^a}{\beta_0}\right) = a \log q + B_0. \quad (\text{A.21})$$

Substitution of (A.19)–(A.21) into (A.12) gives the first inequality of (A.18), with every confidence and logarithmic term retained.

Since $q \geq 1$, $b = 2a + 2$, and $T = q^b + u(L)$,

$$q^{2a} \leq q^b \leq T, \quad a \log q \leq a q^b \leq aT, \quad q^b \geq 1. \quad (\text{A.22})$$

The five summands are controlled separately:

$$q^{2a}T \leq q^bT \leq q^b(1 + T), \quad (\text{A.23})$$

$$a q^{2a} \log q \leq a q^{2a}T \leq a q^bT \leq a q^b(1 + T), \quad (\text{A.24})$$

$$B_0 q^{2a} \leq B_0 q^b \leq B_0 q^b(1 + T), \quad (\text{A.25})$$

$$\alpha_0^{-1} a \log q \leq \alpha_0^{-1} a q^b \leq \alpha_0^{-1} a q^b(1 + T), \quad (\text{A.26})$$

and

$$\alpha_0^{-1} B_0 \leq \alpha_0^{-1} B_0 q^b \leq \alpha_0^{-1} B_0 q^b(1 + T). \quad (\text{A.27})$$

Thus

$$q^{2a}(T + a \log q + B_0) + \alpha_0^{-1}(a \log q + B_0) \leq D_a q^b(1 + T). \quad (\text{A.28})$$

Using (A.17) and $\varepsilon_0^{-1} \leq \varepsilon_0^{-2}$ gives

$$\frac{C_{\text{BLM}}}{\varepsilon_0} D_a q^b(1 + T) \leq K_a \varepsilon_0^{-2} q^b(1 + T). \quad (\text{A.29})$$

The integer n_C is at most this real quantity and hence at most its ceiling $N_a(q, L, \varepsilon_0)$.

At $q = 1$, one has $m_C = 1$, $q^{-a} = 1$, $\log q = 0$, $q^{2a} = q^b = 1$, and $T \geq 1$, so (A.19)–(A.29) remain literal. No division by $\log q$ or assumption $q > 1$ occurs. The inequality $\varepsilon_0^{-1} \leq \varepsilon_0^{-2}$ holds throughout $(0, 1)$ and tends to equality as $\varepsilon_0 \uparrow 1$, so K_a has no dependence on a gap from 1. $\square$

Proposition A.3 (Exact-size padding preserves utility and privacy). *Under Assumption 1, Proposition A.2, and Lemma A.3, fix a domain X , a nonempty finite $C \subseteq \{0, 1\}^X$, and $\varepsilon_0 \in (0, 1)$. Put $q = q(C)$, $L = \log |C|$, and*

$$N = N_a(q, L, \varepsilon_0).$$

Then there is a kernel

$$M_C : Z_X^N \rightsquigarrow \mathcal{H}_X$$

with exact input arity N , arbitrary output, distribution-free realizable (α_0, β_0) -PAC utility, and $(\varepsilon_0, \delta_a(q, L))$ -differential privacy. No properness, representation, or computational restriction is introduced.

Proof. Lemma A.3 gives $n_C \leq N$. Let

$$\pi_{n_C}^N(z_1, \dots, z_N) = (z_1, \dots, z_{n_C}) \quad (\text{A.30})$$

and define

$$M_C = A_C \circ \pi_{n_C}^N. \quad (\text{A.31})$$

Thus M_C accepts exactly N records and ignores the final $N - n_C$. If $N = n_C$, it equals A_C .

For every distribution Q realizable by C , the prefix of a sample $S \sim Q^N$ has law Q^{n_C} . Therefore the full output laws, including internal randomness, agree and

$$\begin{aligned} \Pr_{S \sim Q^N, M_C} [R_Q(M_C(S)) > \alpha_0] &= \Pr_{S_0 \sim Q^{n_C}, A_C} [R_Q(A_C(S_0)) > \alpha_0] \\ &\leq \beta_0. \end{aligned} \quad (\text{A.32})$$

The kernel, its arity, and the public projection are fixed before Q , so this is distribution-free utility.

If $S, S' \in Z_X^N$ are replacement-adjacent, projection cannot increase the number of differing records:

$$\#\{i \leq n_C : S_i \neq S'_i\} \leq \#\{i \leq N : S_i \neq S'_i\} \leq 1. \quad (\text{A.33})$$

For every measurable $E \subseteq \mathcal{H}_X$,

$$\begin{aligned} \Pr[M_C(S) \in E] &= \Pr[A_C(\pi_{n_C}^N(S)) \in E] \\ &\leq e^{\varepsilon_0} \Pr[A_C(\pi_{n_C}^N(S')) \in E] + \delta_a(q(C), \log |C|) \\ &= e^{\varepsilon_0} \Pr[M_C(S') \in E] + \delta_a(q(C), \log |C|). \end{aligned} \quad (\text{A.34})$$

Interchanging S, S' gives the reverse inequality. If the changed record is in the ignored suffix, the two prefixes and hence the two output laws are identical; if it lies in the prefix, (A.33) invokes the original replacement comparison. Padding therefore has zero utility and privacy residual. Combining (A.18) with this projection establishes the exact sample-size and learner interface used below. $\square$

A.3 The Exact Privacy Schedule

Lemma A.4 (The cardinality correction). *For every $L \geq 0$, the quantity*

$$u(L) = \log \log(e^e + L)$$

is well-defined and satisfies

$$u(L) \geq 1, \quad e^{-u(L)} = \frac{1}{\log(e^e + L)}. \quad (\text{A.35})$$

For every $R > 0$,

$$L \geq \exp(\exp R) \implies u(L) \geq R. \quad (\text{A.36})$$

Consequently $u(L) \rightarrow \infty$ as $L \rightarrow \infty$. At $L = 0$, $u(0) = 1$ and $e^{-u(0)} = e^{-1}$.

Proof. For $L \geq 0$,

$$e^e + L \geq e^e > 1, \quad \log(e^e + L) \geq \log(e^e) = e > 0. \quad (\text{A.37})$$

Thus both logarithms are defined, and monotonicity gives

$$u(L) = \log \log(e^e + L) \geq \log e = 1. \quad (\text{A.38})$$

Exact exponentiation yields

$$e^{-u(L)} = \exp[-\log \log(e^e + L)] = \frac{1}{\log(e^e + L)}. \quad (\text{A.39})$$

If $L \geq \exp(\exp R)$, then

$$u(L) \geq \log \log(\exp(\exp R)) = R, \quad (\text{A.40})$$

which is the quantified divergence statement. Finally,

$$u(0) = \log \log(e^e) = 1, \quad e^{-u(0)} = e^{-1} = \frac{1}{\log(e^e)}. \quad (\text{A.41})$$

□

Proposition A.4 (Exact privacy logarithm). *Under Assumption 1, Lemma A.4, and Proposition A.3, let $a \geq 1$ be the universal integer in the assumption and put $b = 2a + 2$. For every $q \geq 1$ and $L \geq 0$,*

$$\delta_a(q, L) = \exp[-q^b - \log \log(e^e + L)] = \frac{e^{-q^b}}{\log(e^e + L)} \in (0, e^{-2}], \quad (\text{A.42})$$

and

$$\log \frac{1}{\delta_a(q, L)} = T_a(q, L) = q^{2a+2} + \log \log(e^e + L). \quad (\text{A.43})$$

Moreover, for every domain X , every nonempty finite $C \subseteq \{0, 1\}^X$, and every $\varepsilon_0 \in (0, 1)$, set $q = q(C)$ and $L = \log |C|$. The learner M_C furnished by Proposition A.3 is $(\varepsilon_0, \delta_a(q, L))$ -differentially private, where the parameter and its logarithm are exactly the two preceding displays. At $L = 0$ the privacy logarithm is $q^b + 1$; at $q = 1$ it is $1 + \log \log(e^e + L)$; and at $(q, L) = (1, 0)$ it is 2.

Proof. Since $a \geq 1$,

$$b = 2a + 2 \geq 4 > 0. \quad (\text{A.44})$$

Thus $q^b \geq 1$, while Lemma A.4 gives $u(L) \geq 1$. Hence

$$T_a(q, L) = q^b + u(L) \geq 2 \quad (\text{A.45})$$

and

$$0 < \delta_a(q, L) = e^{-T_a(q, L)} \leq e^{-2} < 1. \quad (\text{A.46})$$

Using (A.39),

$$\delta_a(q, L) = e^{-q^b} e^{-u(L)} = \frac{e^{-q^b}}{\log(e^e + L)}, \quad (\text{A.47})$$

which proves (A.42). Positivity permits reciprocation, so

$$\begin{aligned} \log \frac{1}{\delta_a(q, L)} &= \log(e^{T_a(q, L)}) \\ &= T_a(q, L) = q^b + u(L) \\ &= q^{2a+2} + \log \log(e^e + L). \end{aligned} \quad (\text{A.48})$$

This is an equality, not an asymptotic simplification.

For the class specialization, Proposition A.3 gives $(\varepsilon_0, \delta_a(q(C), \log |C|))$ -privacy. Substituting $q = q(C)$ and $L = \log |C|$ into the identical quantity from (A.42) changes neither the kernel nor the privacy allowance. At $L = 0$,

$$\delta_a(q, 0) = e^{-q^b-1}, \quad \log \frac{1}{\delta_a(q, 0)} = q^b + 1. \quad (\text{A.49})$$

At $q = 1$,

$$\delta_a(1, L) = \frac{e^{-1}}{\log(e^e + L)}, \quad \log \frac{1}{\delta_a(1, L)} = 1 + \log \log(e^e + L). \quad (\text{A.50})$$

These yield $\delta_a(1, 0) = e^{-2}$, with no requirement that $q > 1$ or $L > 0$. $\square$

Proposition A.5 (Activation on bounded-complexity sequences). *Under Proposition A.4, let $a \geq 1$ be the universal integer from Assumption 1, put $b = 2a + 2$, and let $(q_\kappa, L_\kappa)_{\kappa \in \mathbb{N}}$ satisfy, for every κ ,*

$$L_\kappa \geq 0, \quad 1 \leq q_\kappa \leq \bar{q}, \quad (\text{A.51})$$

where $\bar{q} \geq 1$ is finite, and suppose $L_\kappa \rightarrow \infty$. Then

$$\delta_a(q_\kappa, L_\kappa) = \frac{e^{-q_\kappa^b}}{\log(e^e + L_\kappa)} \rightarrow 0, \quad (\text{A.52})$$

even though q_κ need not converge. More precisely,

$$e^{-\bar{q}^b} \leq e^{-q_\kappa^b} \leq e^{-1}, \quad (\text{A.53})$$

so the cardinality factor, rather than the polynomial-in- q_κ factor, forces the decay. In particular,

$$\delta_a(1, L_\kappa) = \frac{e^{-1}}{\log(e^e + L_\kappa)} \rightarrow 0. \quad (\text{A.54})$$

Proof. From (A.51) and $b > 0$, $1 \leq q_\kappa^b \leq \bar{q}^b$. Since $x \mapsto e^{-x}$ is decreasing, this proves (A.53). Thus a schedule containing only $e^{-q_\kappa^b}$ would stay at least $e^{-\bar{q}^b} > 0$ on this sequence.

In contrast, Lemma A.4 gives

$$u(L_\kappa) \rightarrow \infty, \quad e^{-u(L_\kappa)} = \frac{1}{\log(e^e + L_\kappa)} \rightarrow 0. \quad (\text{A.55})$$

Using (A.47) and the upper bound in (A.53),

$$0 < \delta_a(q_\kappa, L_\kappa) \leq \frac{e^{-1}}{\log(e^e + L_\kappa)} \rightarrow 0. \quad (\text{A.56})$$

Equation (A.54) is the exact specialization $q_\kappa = 1$. Accordingly, the explicit cardinality correction preserves privacy decay in the bounded- q_κ regime without assuming $q_\kappa \rightarrow \infty$. $\square$

A.4 Subpower Consequences of Superpolynomial Separation

Throughout this subsection, let $\{C_\kappa\}$ be a sequence of nonempty finite classes satisfying $|C_\kappa| \rightarrow \infty$, the counting bound

$$0 \leq \ell_\kappa \leq \log_2 |C_\kappa| = \frac{L_\kappa}{\log 2},$$

and

$$\forall p \in \mathbb{N} \exists \kappa_0(p) \forall \kappa \geq \kappa_0(p) : \quad L_\kappa > d_\kappa^p. \quad (\text{A.57})$$

Then $L_\kappa = \log |C_\kappa| \rightarrow \infty$, so all subsequent eventual bounds may be restricted to $L_\kappa \geq 1$.

Lemma A.5 (Every-exponent control of the iterated-log term). *Let $\{C_\kappa\}_{\kappa \in \mathbb{N}}$ be a sequence of nonempty finite binary classes, and define $L_\kappa = \log |C_\kappa|$, $\ell_\kappa = \text{LD}(C_\kappa)$, and $s_\kappa = 1 + \log^*(1 + \ell_\kappa)$. Suppose $|C_\kappa| \rightarrow \infty$ and $0 \leq \ell_\kappa \leq L_\kappa / \log 2$ for every κ . Then, for every $r > 0$, there is $\kappa_s(r)$ such that*

$$\kappa \geq \kappa_s(r) \implies s_\kappa = 1 + \log^*(1 + \ell_\kappa) \leq L_\kappa^r. \quad (\text{A.58})$$

This includes $\ell_\kappa = 0$, for which $s_\kappa = 1$.

Proof. The stopping time $x \mapsto \log^* x$ is nondecreasing on $[1, \infty)$. Indeed, if $1 \leq x \leq y$ and $m = \log^* y$, then either the iterates of x reach a value at most 1 before time m , or monotonicity of $\log_2$ through the active iterates gives $\log_2^{(m)} x \leq \log_2^{(m)} y \leq 1$.

For each integer $n \geq 1$,

$$\log^*(2^n) = 1 + \log^* n \leq n. \quad (\text{A.59})$$

The claim holds for $n = 1$. If it holds at $n - 1$ and $n \geq 2$, then $n \leq 2^{n-1}$, so monotonicity gives $\log^* n \leq \log^*(2^{n-1}) \leq n - 1$. For $x > 1$, taking $n = \lceil \log_2 x \rceil$ yields

$$\log^* x \leq n \leq 1 + \log_2 x. \quad (\text{A.60})$$

At $x = 1$, $\log^* 1 = 0$, so the same inequality holds.

When $L_\kappa \geq 1$,

$$1 + \ell_\kappa \leq 1 + \frac{L_\kappa}{\log 2} \leq c_2 L_\kappa, \quad c_2 = 1 + \frac{1}{\log 2}. \quad (\text{A.61})$$

Using (A.60),

$$\begin{aligned} s_\kappa &\leq 2 + \log_2(1 + \ell_\kappa) \\ &\leq 2 + \log_2(c_2 L_\kappa) = A_0 + \frac{\log L_\kappa}{\log 2}, \quad A_0 = 2 + \frac{\log c_2}{\log 2}. \end{aligned} \quad (\text{A.62})$$

Fix $r > 0$. For $L \geq 1$,

$$\log L = \int_1^L \frac{dx}{x} \leq \int_1^L x^{r/2-1} dx = \frac{2}{r}(L^{r/2} - 1) \leq \frac{2}{r}L^{r/2}. \quad (\text{A.63})$$

Thus

$$s_\kappa \leq A_r L_\kappa^{r/2}, \quad A_r = A_0 + \frac{2}{r \log 2}. \quad (\text{A.64})$$

Since $L_\kappa \rightarrow \infty$, eventually $L_\kappa^{r/2} \geq A_r$, and then $A_r L_\kappa^{r/2} \leq L_\kappa^r$. If $\ell_\kappa = 0$, then $\log^* 1 = 0$ and the same eventual bound holds without any division by ℓ_κ . $\square$

Proposition A.6 (Superpolynomial separation implies subpower total complexity). *Let $\{C_\kappa\}_{\kappa \in \mathbb{N}}$ be a sequence of nonempty finite binary classes. Define*

$$L_\kappa = \log |C_\kappa|, \quad d_\kappa = \text{VC}(C_\kappa), \quad \ell_\kappa = \text{LD}(C_\kappa), \quad s_\kappa = 1 + \log^*(1 + \ell_\kappa), \quad q_\kappa = d_\kappa + s_\kappa.$$

Suppose $|C_\kappa| \rightarrow \infty$, $0 \leq \ell_\kappa \leq L_\kappa / \log 2$ for every κ , and

$$\forall p \in \mathbb{N} \ \exists \kappa_0(p) \ \forall \kappa \geq \kappa_0(p) : \quad L_\kappa > d_\kappa^p.$$

Then, under Lemma A.5,

$$\forall t > 0 \ \exists \kappa_q(t) \ \forall \kappa \geq \kappa_q(t) : \quad q_\kappa \leq L_\kappa^t. \quad (\text{A.65})$$

Consequently $q_\kappa = L_\kappa^{o(1)}$. The conclusion permits $d_\kappa = 0$ and bounded or nonconvergent q_κ .

Proof. Fix $t > 0$ and choose one integer

$$p = p(t) > \frac{2}{t}, \quad \frac{1}{p} < \frac{t}{2}. \quad (\text{A.66})$$

Condition (A.57) gives, for every sufficiently large κ ,

$$L_\kappa > d_\kappa^p. \quad (\text{A.67})$$

After also requiring $L_\kappa \geq 1$, taking the positive p -th root gives

$$d_\kappa < L_\kappa^{1/p} \leq L_\kappa^{t/2}. \quad (\text{A.68})$$

This remains valid at $d_\kappa = 0$. Applying Lemma A.5 with exponent $t/2$ gives

$$s_\kappa \leq L_\kappa^{t/2} \quad (\text{A.69})$$

on another tail. After additionally requiring $L_\kappa^{t/2} \geq 2$,

$$q_\kappa = d_\kappa + s_\kappa < 2L_\kappa^{t/2} \leq L_\kappa^t. \quad (\text{A.70})$$

The threshold is the maximum of finitely many thresholds and therefore applies to every later index. Since $q_\kappa \geq 1$ and eventually $L_\kappa > 1$, (A.65) also gives

$$0 \leq \frac{\log q_\kappa}{\log L_\kappa} \leq t \quad \text{eventually.} \quad (\text{A.71})$$

Letting the arbitrary t decrease to zero proves the $L_\kappa^{o(1)}$ statement. $\square$

Lemma A.6 (Subpower growth of the privacy schedule). *Under Assumption 1, Lemma A.4, Proposition A.4, and Proposition A.6, let $a \geq 1$ be the universal integer in the assumption, put $b = 2a + 2$, and let L_κ, q_κ be the sequence quantities defined in Proposition A.6. For every $r > 0$ there is $\kappa_T(r)$ such that*

$$\kappa \geq \kappa_T(r) \implies u(L_\kappa) \leq L_\kappa^r, \quad T_a(q_\kappa, L_\kappa) \leq L_\kappa^r. \quad (\text{A.72})$$

Hence $u(L_\kappa) = L_\kappa^{o(1)}$ and $T_a(q_\kappa, L_\kappa) = L_\kappa^{o(1)}$, including when q_κ is bounded.

Proof. For $L \geq 1$, put $z = \log(e^e + L) > 1$. Then

$$\log z = \int_1^z \frac{dx}{x} \leq z - 1 \leq z.$$

Also $e^e + L \leq (e^e + 1)L$, so

$$u(L) \leq \log(e^e + L) \leq \log(e^e + 1) + \log L. \quad (\text{A.73})$$

Fix $r > 0$. Applying (A.63) gives

$$u(L_\kappa) \leq B_r L_\kappa^{r/2}, \quad B_r = \log(e^e + 1) + \frac{2}{r}. \quad (\text{A.74})$$

Eventually $L_\kappa^{r/2} \geq B_r$, hence

$$u(L_\kappa) \leq L_\kappa^r. \quad (\text{A.75})$$

Now fix a fresh target exponent $r > 0$. Proposition A.6, used at $r/(2b) > 0$, gives

$$q_\kappa^b \leq L_\kappa^{r/2} \quad (\text{A.76})$$

eventually. The first part, used at exponent $r/2$, gives

$$u(L_\kappa) \leq L_\kappa^{r/2} \quad (\text{A.77})$$

eventually. On a common tail with $L_\kappa^{r/2} \geq 2$,

$$T_a(q_\kappa, L_\kappa) = q_\kappa^b + u(L_\kappa) \leq 2L_\kappa^{r/2} \leq L_\kappa^r. \quad (\text{A.78})$$

The lower bounds $u(L_\kappa) \geq 1$ and $T_a(q_\kappa, L_\kappa) \geq 2$, together with (A.72), yield

$$\frac{\log u(L_\kappa)}{\log L_\kappa} \rightarrow 0, \quad \frac{\log T_a(q_\kappa, L_\kappa)}{\log L_\kappa} \rightarrow 0. \quad (\text{A.79})$$

If $q_\kappa \leq \bar{q}$ on a tail, then directly

$$T_a(q_\kappa, L_\kappa) \leq \bar{q}^b + u(L_\kappa), \quad (\text{A.80})$$

and the fixed first term and (A.75) are absorbed by every positive power of L_κ . $\square$

Proposition A.7 (The exact sample size is subpower and sublogarithmic). *Under Assumption 1, Lemma A.3, Proposition A.6, and Lemma A.6, fix $\varepsilon_0 \in (0, 1)$, let K_a be the constant fixed by Lemma A.3, and let L_κ, q_κ be the sequence quantities in Proposition A.6. Define*

$$N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0).$$

Then

$$\forall r > 0 \ \exists \kappa_N(r) \ \forall \kappa \geq \kappa_N(r) : \quad N_\kappa \leq L_\kappa^r. \quad (\text{A.81})$$

Consequently

$$N_\kappa = L_\kappa^{o(1)}, \quad N_\kappa = o(L_\kappa). \quad (\text{A.82})$$

Both conclusions remain true for bounded q_κ , including $q_\kappa \equiv 1$.

Proof. Set

$$H = K_a \varepsilon_0^{-2}. \quad (\text{A.83})$$

By Lemma A.3, H is finite and fixed while $\kappa \rightarrow \infty$, with dependence only on $a, \alpha_0, \beta_0, C_{\text{BLM}}, \varepsilon_0$.

Fix $r > 0$. Proposition A.6, at exponent $r/(4b)$, and Lemma A.6, at $r/4$, give

$$q_\kappa^b \leq L_\kappa^{r/4}, \quad T_a(q_\kappa, L_\kappa) \leq L_\kappa^{r/4} \quad (\text{A.84})$$

on a common tail. When $L_\kappa \geq 1$,

$$\begin{aligned} q_\kappa^b(1 + T_a(q_\kappa, L_\kappa)) &\leq L_\kappa^{r/4}(1 + L_\kappa^{r/4}) \\ &\leq 2L_\kappa^{r/2}. \end{aligned} \quad (\text{A.85})$$

Using the exact ceiling and $\lceil x \rceil \leq x + 1$,

$$\begin{aligned} N_\kappa &= \left\lceil H q_\kappa^b(1 + T_a(q_\kappa, L_\kappa)) \right\rceil \\ &\leq 2H L_\kappa^{r/2} + 1 \\ &\leq (2H + 1) L_\kappa^{r/2}. \end{aligned} \quad (\text{A.86})$$

Eventually $L_\kappa^{r/2} \geq 2H + 1$, so

$$N_\kappa \leq L_\kappa^r. \quad (\text{A.87})$$

Thus the fixed power, sum, product, prefactor, additive unit, and ceiling have each been explicitly controlled.

Because $N_\kappa \geq 1$ and eventually $L_\kappa > 1$,

$$0 \leq \frac{\log N_\kappa}{\log L_\kappa} \leq r \quad (\text{A.88})$$

eventually for every $r > 0$, proving the subpower statement. Taking $r = 1/2$ in (A.81) gives

$$0 \leq \frac{N_\kappa}{L_\kappa} \leq L_\kappa^{-1/2} \longrightarrow 0, \quad (\text{A.89})$$

which proves $N_\kappa = o(L_\kappa)$ directly.

If $q_\kappa \leq \bar{q}$ on a tail, (A.80) and the exact sample formula yield

$$N_\kappa \leq 1 + H\bar{q}^b(1 + \bar{q}^b + u(L_\kappa)). \quad (\text{A.90})$$

Using (A.75) at exponent $1/2$,

$$\frac{N_\kappa}{L_\kappa} \leq \frac{1 + H\bar{q}^b(1 + \bar{q}^b)}{L_\kappa} + H\bar{q}^b L_\kappa^{-1/2} \longrightarrow 0. \quad (\text{A.91})$$

This retains the bounded-complexity boundary. Combining Proposition A.6, Lemma A.6, and the present result gives

$$T_a(q_\kappa, L_\kappa) = L_\kappa^{o(1)}, \quad N_\kappa = L_\kappa^{o(1)} = o(L_\kappa). \quad (\text{A.92})$$

□

A.5 Uniform Privacy Negligibility

Lemma A.7 (Uniform ceiling control). *Under Assumption 1 and Lemma A.4, let $a \geq 1$ be the universal integer in the assumption, put $b = 2a + 2$, fix $\varepsilon_0 \in (0, 1)$, and let $K_a \geq 2$ be the constant used in the pointwise definition of N_a . Let $\{X_\kappa\}_{\kappa \in \mathbb{N}}$ be any sequence of domains and, for every κ , let $C_\kappa \subseteq \{0, 1\}^{X_\kappa}$ be nonempty and finite. Put $L_\kappa = \log |C_\kappa|$ and $q_\kappa = q(C_\kappa)$, and define*

$$x_\kappa = q_\kappa^b, \quad u_\kappa = u(L_\kappa), \quad A_\kappa = K_a \varepsilon_0^{-2} x_\kappa (1 + x_\kappa + u_\kappa).$$

Also define $N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0)$. Then, for every κ ,

$$A_\kappa \geq 6, \quad N_\kappa = \lceil A_\kappa \rceil \geq 1, \quad N_\kappa \leq 2A_\kappa. \quad (\text{A.93})$$

Proof. The basic setup gives $q_\kappa \geq 1$, hence $x_\kappa = q_\kappa^b \geq 1$. Lemma A.4 gives $u_\kappa \geq 1$, so $1 + x_\kappa + u_\kappa \geq 3$. The pointwise definition has $K_a \geq 2$, and $\varepsilon_0^{-2} \geq 1$. Therefore

$$A_\kappa \geq 2 \cdot 1 \cdot 1 \cdot 3 = 6.$$

Indeed, the pointwise definition of N_a and $T_a(q_\kappa, L_\kappa) = x_\kappa + u_\kappa$ give

$$N_\kappa = \lceil K_a \varepsilon_0^{-2} x_\kappa (1 + x_\kappa + u_\kappa) \rceil = \lceil A_\kappa \rceil.$$

Consequently

$$N_\kappa \leq A_\kappa + 1 \leq A_\kappa + \frac{1}{6} A_\kappa \leq 2A_\kappa.$$

The argument is pointwise and imposes no convergence condition on q_κ . □

Lemma A.8 (Two-factor privacy decomposition). *Under Proposition A.4, Lemma A.7, and a fixed $\rho > 0$, let $q_\kappa, L_\kappa, x_\kappa, u_\kappa, N_\kappa$ be the exact sequence quantities defined in Lemma A.7, set $\delta_\kappa = \delta_a(q_\kappa, L_\kappa)$, and define*

$$C_{\rho, \varepsilon_0, K_a} = (2K_a \varepsilon_0^{-2})^\rho, \\ F_\rho(x) = e^{-x} x^\rho (1 + x)^\rho, \quad H_\rho(u) = e^{-u} (1 + u)^\rho.$$

Then, for every κ ,

$$\delta_\kappa N_\kappa^\rho \leq C_{\rho, \varepsilon_0, K_a} F_\rho(x_\kappa) H_\rho(u_\kappa). \quad (\text{A.94})$$

Proof. Proposition A.4 gives $\delta_\kappa = e^{-x_\kappa - u_\kappa}$, while Lemma A.7 gives

$$N_\kappa^\rho \leq [2K_a \varepsilon_0^{-2} x_\kappa (1 + x_\kappa + u_\kappa)]^\rho.$$

For nonnegative x, u ,

$$1 + x + u \leq (1 + x)(1 + u), \quad (\text{A.95})$$

because the right side is $1 + x + u + xu$. Since $\rho > 0$, raising positive quantities to ρ preserves order, and hence

$$\begin{aligned} \delta_\kappa N_\kappa^\rho &\leq (2K_a \varepsilon_0^{-2})^\rho e^{-x_\kappa - u_\kappa} [x_\kappa (1 + x_\kappa + u_\kappa)]^\rho \\ &\leq (2K_a \varepsilon_0^{-2})^\rho [e^{-x_\kappa} x_\kappa^\rho (1 + x_\kappa)^\rho] [e^{-u_\kappa} (1 + u_\kappa)^\rho]. \end{aligned} \quad (\text{A.96})$$

This is (A.94) for the exact N_κ and δ_κ , with no asymptotic proxy. $\square$

Lemma A.9 (Uniform exponential-polynomial bound). *For every fixed $\rho > 0$, define $F_\rho(x) = e^{-x} x^\rho (1 + x)^\rho$ for $x \geq 1$. Then*

$$\sup_{x \geq 1} F_\rho(x) \leq B_\rho := 2^\rho e^{-x_\rho} x_\rho^{2\rho} < \infty, \quad x_\rho = \max\{1, 2\rho\}. \quad (\text{A.97})$$

Proof. For $x \geq 1$, $1 + x \leq 2x$, so

$$F_\rho(x) \leq 2^\rho e^{-x} x^{2\rho}. \quad (\text{A.98})$$

Let $g_\rho(x) = e^{-x} x^{2\rho}$. On $(0, \infty)$,

$$g'_\rho(x) = e^{-x} x^{2\rho-1} (2\rho - x). \quad (\text{A.99})$$

If $2\rho \leq 1$, the function is nonincreasing on $[1, \infty)$, so its maximum there is $g_\rho(1)$. If $2\rho > 1$, it increases on $[1, 2\rho]$ and decreases afterward, so its maximum is $g_\rho(2\rho)$. In both cases the maximizer is x_ρ , and (A.98) gives (A.97). The constant depends only on the fixed ρ , not on the sequence. $\square$

Proposition A.8 (Privacy negligibility). *Under Assumption 1, Lemma A.4, Proposition A.4, and Lemmas A.7–A.9, fix $\varepsilon_0 \in (0, 1)$, let $K_a \geq 2$ be the constant used in the pointwise definition of N_a , let $\{X_\kappa\}_{\kappa \in \mathbb{N}}$ be any sequence of domains, and let $\{C_\kappa\}_{\kappa \in \mathbb{N}}$ satisfy $C_\kappa \subseteq \{0, 1\}^{X_\kappa}$ nonempty and finite and $|C_\kappa| \rightarrow \infty$. Put*

$$L_\kappa = \log |C_\kappa|, \quad q_\kappa = q(C_\kappa),$$

and define

$$\delta_\kappa = \delta_a(q_\kappa, L_\kappa), \quad N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0).$$

For every fixed $\rho > 0$,

$$\begin{aligned} \forall \tau > 0 \quad \exists \kappa_0(\rho, \tau) \quad \forall \kappa \geq \kappa_0(\rho, \tau) : \\ \delta_\kappa N_\kappa^\rho \leq \tau. \end{aligned} \quad (\text{A.100})$$

Equivalently,

$$\delta_\kappa N_\kappa^\rho \longrightarrow 0 \quad \text{for every fixed } \rho > 0. \quad (\text{A.101})$$

The conclusion is uniform over all possible behavior of $q_\kappa \geq 1$, including bounded, constant, oscillatory, and unbounded sequences. No superpolynomial separation condition on d_κ is required.

Proof. Size divergence gives

$$L_\kappa = \log |C_\kappa| \longrightarrow \infty, \quad (\text{A.102})$$

and Lemma A.4 then gives

$$u_\kappa = u(L_\kappa) \longrightarrow \infty. \quad (\text{A.103})$$

For $u \geq 1$,

$$\log u = \int_1^u \frac{dt}{t} \leq \int_1^u \frac{dt}{\sqrt{t}} = 2(\sqrt{u} - 1) \leq 2\sqrt{u}.$$

Since $1 + u \leq 2u$,

$$\log(1 + u) \leq \log 2 + \log u \leq 3\sqrt{u}. \quad (\text{A.104})$$

Whenever $u \geq 36\rho^2$,

$$\rho \log(1 + u) \leq 3\rho\sqrt{u} \leq \frac{u}{2}, \quad H_\rho(u) \leq e^{-u/2}. \quad (\text{A.105})$$

Let

$$D_\rho = C_{\rho, \varepsilon_0, K_a} B_\rho.$$

By (A.94) and (A.97),

$$\delta_\kappa N_\kappa^\rho \leq D_\rho H_\rho(u_\kappa) \leq D_\rho e^{-u_\kappa/2} \quad (\text{A.106})$$

whenever $u_\kappa \geq 36\rho^2$. For $\tau > 0$, define

$$U_{\rho, \tau} = \max \left\{ 1, 36\rho^2, 2 \log \frac{D_\rho}{\tau} \right\}. \quad (\text{A.107})$$

Because $u_\kappa \rightarrow \infty$, eventually $u_\kappa \geq U_{\rho, \tau}$, and then $D_\rho e^{-u_\kappa/2} \leq \tau$. If $D_\rho/\tau < 1$, the logarithmic entry is negative and the desired inequality is already automatic for $u_\kappa \geq 0$.

The factor $F_\rho(x_\kappa)$ was bounded uniformly on the entire domain $x_\kappa = q_\kappa^b \geq 1$. Thus if q_κ is bounded, constant, nonconvergent, or oscillatory, the same bound applies while the independent u_κ -factor vanishes. If $q_\kappa \rightarrow \infty$, exponential decay in x_κ can only improve the estimate. $\square$

Proposition A.9 (Standard approximate-privacy allowance). *Under Lemma A.7 and Proposition A.8, let N_κ, δ_κ be the exact quantities defined in those results. For every fixed $c > 0$ there is κ_c such that*

$$\kappa \geq \kappa_c \implies \delta_\kappa \leq \frac{c}{N_\kappa^2 \log(eN_\kappa)}. \quad (\text{A.108})$$

Proof. Apply Proposition A.8 at the fixed exponent $\rho = 3$ and tolerance c . Eventually,

$$\delta_\kappa N_\kappa^3 \leq c. \quad (\text{A.109})$$

Lemma A.7 gives $N_\kappa \geq 1$. For every real $N \geq 1$,

$$\log(eN) = 1 + \log N \leq N, \quad (\text{A.110})$$

because $f(N) = N - 1 - \log N$ satisfies $f(1) = 0$ and $f'(N) = 1 - N^{-1} \geq 0$. Hence, on the tail in (A.109),

$$\delta_\kappa \leq \frac{c}{N_\kappa^3} \leq \frac{c}{N_\kappa^2 \log(eN_\kappa)}.$$

The choice $\rho = 3$ is made only after the every-fixed- ρ result has been proved, and no growth assumption on q_κ is used. $\square$

A.6 Sequence Assembly and the Lower-Bound Contradiction

Proposition A.10 (Exact private learner family). *Under Assumption 1, Lemma A.3, Proposition A.3, and Proposition A.4, fix $\alpha_0 = \beta_0 = 1/8$, $\varepsilon_0 \in (0, 1)$, any sequence of domains $\{X_\kappa\}_{\kappa \in \mathbb{N}}$, and any sequence $\{C_\kappa\}_{\kappa \in \mathbb{N}}$, with $C_\kappa \subseteq \{0, 1\}^{X_\kappa}$ nonempty and finite. Put*

$$L_\kappa = \log |C_\kappa|, \quad d_\kappa = d(C_\kappa), \quad q_\kappa = q(C_\kappa).$$

Suppose $|C_\kappa| \rightarrow \infty$ and

$$\forall p \in \mathbb{N} \exists \kappa_0(p) \forall \kappa \geq \kappa_0(p) : \quad L_\kappa > d_\kappa^p.$$

Then there is one constant K_a , depending only on $a, \alpha_0, \beta_0, C_{\text{BLM}}$, and for every κ there is a kernel

$$M_\kappa : Z_{X_\kappa}^{N_\kappa} \rightsquigarrow \mathcal{H}_{X_\kappa}, \quad N_\kappa = N_a(q_\kappa, L_\kappa, \varepsilon_0), \quad (\text{A.111})$$

such that:

(i) M_κ has exact input arity N_κ , arbitrary output, and no representation or computational restriction;

(ii) for every Q realizable by C_κ ,

$$\Pr_{S \sim Q^{N_\kappa, M_\kappa}} [R_Q(M_\kappa(S)) > \alpha_0] \leq \beta_0; \quad (\text{A.112})$$

(iii) M_κ is $(\varepsilon_0, \delta_\kappa)$ -differentially private, where

$$\delta_\kappa = \exp[-q_\kappa^{2a+2} - \log \log(e^e + L_\kappa)]; \quad (\text{A.113})$$

(iv)

$$\log \frac{1}{\delta_\kappa} = q_\kappa^{2a+2} + \log \log(e^e + L_\kappa), \quad (\text{A.114})$$

and

$$\begin{aligned} N_\kappa &= \lceil K_a \varepsilon_0^{-2} q_\kappa^{2a+2} (1 + q_\kappa^{2a+2} + \log \log(e^e + L_\kappa)) \rceil \\ &\leq 1 + K_a \varepsilon_0^{-2} q_\kappa^{2a+2} (1 + q_\kappa^{2a+2} + \log \log(e^e + L_\kappa)). \end{aligned} \quad (\text{A.115})$$

These conclusions hold at every index, including finite prefixes and the boundary $q_\kappa = 1$.

Proof. Assumption 1 fixes the same universal a before the sequence is considered. Lemma A.3 fixes

$$K_a = \max\{2, C_{\text{BLM}} D_a\},$$

independently of $\kappa, X_\kappa, C_\kappa, L_\kappa, q_\kappa$, every realizable distribution, and ε_0 . For each κ , apply Proposition A.3 to $C = C_\kappa$, $X = X_\kappa$, $q = q_\kappa$, $L = L_\kappa$, and the common ε_0 . It supplies one kernel M_{C_κ} with exact arity $N_a(q_\kappa, L_\kappa, \varepsilon_0)$. Choose one such kernel at each index and set

$$M_\kappa = M_{C_\kappa}. \quad (\text{A.116})$$

This indexwise selection does not couple the internal randomness and does not make the learner depend on Q .

Equation (A.112) and both ordered privacy inequalities are exactly the conclusions of Proposition A.3. Proposition A.4 identifies the same privacy parameter with (A.113) and proves (A.114). Substituting $b = 2a + 2$ and the definition of T_a into N_a gives the equality in (A.115); $\lceil x \rceil \leq x + 1$ gives its upper bound. Since every class is nonempty and finite, $L_\kappa \geq 0$ and $q_\kappa \geq 1$ at every index, so no asymptotic tail is needed for this construction. $\square$

Proposition A.11 (Rate certificates for the same learner family). *Under Proposition A.10, Proposition A.6, Lemma A.6, Proposition A.7, and Proposition A.8, let $q_\kappa, L_\kappa, T_a(q_\kappa, L_\kappa), N_\kappa$, and δ_κ be the exact quantities defined in Proposition A.10. Then*

$$q_\kappa = L_\kappa^{o(1)}, \quad T_a(q_\kappa, L_\kappa) = L_\kappa^{o(1)}, \quad (\text{A.117})$$

$$N_\kappa = L_\kappa^{o(1)}, \quad \frac{N_\kappa}{L_\kappa} \longrightarrow 0, \quad (\text{A.118})$$

and, for every fixed $\rho > 0$,

$$\delta_\kappa N_\kappa^\rho \longrightarrow 0. \quad (\text{A.119})$$

The statements include bounded, constant, or oscillatory q_κ , and finite prefixes do not affect the limits.

Proof. Proposition A.10 uses exactly the quantities $q_\kappa, L_\kappa, T_a, N_a, \delta_a$ appearing in the earlier results, so there is no proxy sample size or privacy parameter. Proposition A.6 and Lemma A.6 give (A.117). Proposition A.7 gives, for every $r > 0$,

$$N_\kappa \leq L_\kappa^r \quad (\text{A.120})$$

on an eventual tail. Taking $r = 1/2$ and using $L_\kappa \rightarrow \infty$ gives

$$0 \leq \frac{N_\kappa}{L_\kappa} \leq L_\kappa^{-1/2} \rightarrow 0. \quad (\text{A.121})$$

Independently of those superpolynomial-separation consequences, Proposition A.8 applies to the exact pair (A.113), (A.115) and gives (A.119). Its ceiling bound comes directly from the pointwise definition of N_a , and its limit uses only $|C_\kappa| \rightarrow \infty$ to obtain $u(L_\kappa) \rightarrow \infty$; it does not consume any of the preceding subpower conclusions.

If $q_\kappa \leq \bar{q}$ on a tail, the direct bound

$$N_\kappa \leq 1 + K_a \varepsilon_0^{-2} \bar{q}^b (1 + \bar{q}^b + u(L_\kappa))$$

has ratio tending to zero by (A.75), while the proof of Proposition A.8 controls the q_κ -factor uniformly and lets $u(L_\kappa) \rightarrow \infty$. This includes $q_\kappa \equiv 1$. $\square$

Lemma A.10 (Privacy relaxation and source-scale eligibility). *Under Proposition A.10, fix κ and let $\bar{\delta}$ be an allowed privacy schedule. If*

$$\bar{\delta}(N_\kappa) \geq \delta_\kappa, \quad (\text{A.122})$$

then the same M_κ is $(\varepsilon_0, \bar{\delta}(N_\kappa))$ -differentially private in both ordered replacement directions. Equality is allowed, and no reverse implication is claimed when $\bar{\delta}(N_\kappa) < \delta_\kappa$.

Moreover, under Proposition A.9, for every fixed $c > 0$, the same learner is eventually

$$\left(\varepsilon_0, \frac{c}{N_\kappa^2 \log(eN_\kappa)} \right)\text{-differentially private.} \quad (\text{A.123})$$

Proof. Let $S, S' \in Z_{X_\kappa}^{N_\kappa}$ be replacement-adjacent and let $E \subseteq \mathcal{H}_{X_\kappa}$ be measurable. Proposition A.10 gives

$$\Pr[M_\kappa(S) \in E] \leq e^{\varepsilon_0} \Pr[M_\kappa(S') \in E] + \delta_\kappa.$$

Using (A.122) in the valid direction gives

$$\Pr[M_\kappa(S) \in E] \leq e^{\varepsilon_0} \Pr[M_\kappa(S') \in E] + \bar{\delta}(N_\kappa). \quad (\text{A.124})$$

The reverse ordered inequality similarly becomes

$$\Pr[M_\kappa(S') \in E] \leq e^{\varepsilon_0} \Pr[M_\kappa(S) \in E] + \bar{\delta}(N_\kappa). \quad (\text{A.125})$$

Thus only the allowed additive slack has increased; the kernel, sample count, output space, risk, and adjacency relation are unchanged.

Proposition A.9 states that, for every fixed $c > 0$, eventually

$$\delta_\kappa \leq \frac{c}{N_\kappa^2 \log(eN_\kappa)}.$$

Substitution into (A.124)–(A.125) gives (A.123). At the exact schedule no tail is needed; if a comparison holds only eventually, the relaxed conclusion holds on exactly that tail. $\square$

Proposition A.12 (No eventual linear lower bound). *Under Proposition A.10, Proposition A.11, and Lemma A.10, consider either the exact schedule δ_κ , or a schedule $\bar{\delta}$ for which*

$$\bar{\delta}(N_\kappa) \geq \delta_\kappa \quad \text{eventually.} \quad (\text{A.126})$$

There do not exist $c_ > 0$ and κ_* such that, for every $\kappa \geq \kappa_*$, every arbitrary-output, computationally unrestricted, distribution-free realizable (α_0, β_0) -PAC learner for C_κ at the relevant privacy schedule must use at least*

$$c_* L_\kappa \quad (\text{A.127})$$

records.

Proof. Fix arbitrary $c_* > 0$ and κ_* . In the exact-schedule case, set $\kappa_{\text{priv}} = 0$; in the comparison-schedule case, let κ_{priv} be a tail index for (A.126). Proposition A.11 gives $N_\kappa/L_\kappa \rightarrow 0$, so there is $\kappa_{\text{rate}}(c_*)$ such that

$$N_\kappa < c_* L_\kappa \quad (\kappa \geq \kappa_{\text{rate}}(c_*)). \quad (\text{A.128})$$

The ratio is well-defined there because $L_\kappa \rightarrow \infty$. Choose

$$\kappa \geq \max\{\kappa_*, \kappa_{\text{priv}}, \kappa_{\text{rate}}(c_*)\}. \quad (\text{A.129})$$

Proposition A.10 supplies an eligible learner M_κ using exactly N_κ records. In the comparison-schedule case, Lemma A.10 makes the same learner eligible. The hypothetical lower bound would give $N_\kappa \geq c_* L_\kappa$, contradicting (A.128). Since c_* and the proposed tail were arbitrary, no positive eventual linear lower-bound constant exists. The standard allowance follows by taking $\bar{\delta}(N_\kappa) = c/[N_\kappa^2 \log(eN_\kappa)]$ and using (A.123). $\square$

A.7 Proof of the Main Theorem

Proof of Theorem 3.1. Assumption 1 and Proposition A.2 give the arbitrary-output stable-selection learner and its exact two-term raw sample bound. Lemma A.3 proves the displayed sample specialization and fixes a common K_a with only the allowed dependence; Proposition A.3 turns the learner into an exact- N_a kernel without changing utility or privacy. Proposition A.4 proves the displayed value of δ_κ and its exact logarithm.

For an arbitrary admissible sequence, Proposition A.6, Lemma A.6, and Proposition A.7 prove

$$q_\kappa = L_\kappa^{o(1)}, \quad T_a(q_\kappa, L_\kappa) = L_\kappa^{o(1)}, \quad N_\kappa = L_\kappa^{o(1)} = o(L_\kappa).$$

Proposition A.8 proves $\delta_\kappa N_\kappa^\rho \rightarrow 0$ for every fixed $\rho > 0$, and Proposition A.9 gives the stated standard approximate-privacy comparison. These privacy conclusions derive their ceiling directly from the pointwise definition of N_a and require only size divergence; the superpolynomial condition is used separately above for the sample-rate conclusions.

Proposition A.10 attaches all pointwise guarantees to one learner M_κ for each index, and Proposition A.11 attaches the exact sequence rates to those same learners. Lemma A.10 proves privacy monotonicity only toward a larger allowed additive parameter, in both ordered adjacency directions. Proposition A.12 then combines the valid learner with $N_\kappa/L_\kappa \rightarrow 0$ and rules out every eventual positive linear lower bound at the exact, standard, or any eventually weaker privacy allowance. The argument never removes Assumption 1; hence the result is conditional exactly as stated. $\square$